LLM-Augmented Event-Study Strategy on US Large-Cap Earnings Calls

MF815 Machine Learning for Finance Final Project
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1.Executive Summary

We test whether GPT-4o, applied to ~1,500 quarterly earnings calls of 50 US large-cap equities (2018–2025), produces signals that add alpha beyond standard technicals. A RAG pipeline retrieves call excerpts under a strict before-date filter and extracts six structured signals per call (sentiment, confidence, certainty, evasion, guidance change, tone shift). Signals are joined to market-model CAR (+2, +11) labels and fed to a refit-based three-arm ablation: BuyHold_EW50, Tech-only RF, and LLM + Tech RF. The LLM + Tech arm beats Tech-only on val 2023 AUC (0.602 vs 0.521) and lifts Sharpe from −0.13 to 0.95 in the volatile 2025 test, while permutation importance attributes 36.6% of pooled out-of-sample lift to the LLM block. Event-level Spearman $\rho = 0.091$ ($p = 0.042$) and Mann-Whitney $p = 0.029$ confirm the ranking edge is not noise. The single load-bearing signal is evasion: dropping it alone removes 72% of the LLM block's contribution.

2.Motivation and Proposal

Motivation

Most price-relevant information in equities is released in language that financial statements have not yet absorbed. Earnings calls are informationally rich events where prices react and continue drifting for weeks (Ball and Brown, 1968; Bernard and Thomas, 1989, 1990). Lexicon-based and FinBERT-style approaches (Tetlock, 2007; Li, 2010; Loughran and McDonald, 2011) compress an entire transcript into one polarity score, discarding speaker role, cross-quarter dynamics, and multi-dimensional decomposition. We propose a Retrieval-Augmented Generation (Lewis et al., 2020; Gao et al., 2023) pipeline on top of GPT-4o that extracts six structured signals per call, joins them with price-based features, and tests via a refit-based three-arm ablation whether the LLM layer adds alpha beyond technicals on 50 US large-caps over 2018–2025.

Proposal

The pipeline runs as five sequential notebooks, engineered around three constraints: **faithfulness** (every signal grounded in a verifiable quote), **no look-ahead** (every transformation fitted on a strictly earlier window), and **cost discipline** (LLM called once per call, embeddings cached). Diagram in **Appendix A.2.1**.

2.1 RAG + LLM signal extraction (Notebook02). The retrieval-augmented generation paradigm — using embeddings to ground a generative model in a small set of relevant documents — was introduced by Lewis et al. (2020) and has since become standard practice for knowledge-intensive NLP (Gao et al., 2023). Each of 1,520 transcripts (median 14k tokens) is parsed into prepared remarks vs Q&A, word-tokenized into 200/50-word chunks, embedded with text-embedding-3-large, and retrieved by cosine similarity over three fixed queries with a before_date < event_date filter. A single GPT-4o call returns six structured signals (sentiment, confidence, certainty, evasion, guidance_change, tone_vs_prior_quarter) plus ≥2 verbatim evidence quotes. Generative LLM is chosen over lexicon/FinBERT because structured-output mode produces all six heterogeneous signals in one call with auditable citations — neither alternative can. Prompts, hyperparameters, and OpenAI usage in **Appendix D**.

2.2 CAR labels via market-model event study (Notebook03). Six structured signals need a target to learn against. We compute cumulative abnormal returns (CAR) under the standard market-model event-study framework (Brown and Warner, 1985; MacKinlay, 1997): for each call, OLS-fit $r_i = \alpha + \beta \cdot r_{SPY} + \varepsilon$ on the estimation window [−120, −20] trading days, then accumulate residuals over four post-event windows. The training target is **CAR (+2, +11)**: positions open on day +2 to skip announcement-day microstructure noise and close at +11 to balance signal-to-noise against window length. The evaluation window is **CAR (+2, +21)** — deliberately longer than the training window, as a stricter PEAD check: a model that learns short-horizon noise will fail on the longer horizon. A **tail filter `$|CAR| > 0.5 \cdot \sigma_e \cdot \sqrt{10}$`** (σ_e from the OLS residual sd) drops events below the noise threshold for *training and validation only* — test events are never filtered, so the backtest in NB05 sees the full sample. The 1,496 LLM-scored sessions become 1,480 valid CAR rows (16 lost to insufficient estimation history), which split into 925/192/198/165 events across train/val/test 2024/test 2025. **Appendix E** holds CAR distributions, sector-level β statistics (Utilities 0.30, NVDA 2.04), and class-balance tables.

2.3 Model bake-off and Optuna-tuned Random Forest (Notebook04). With features and target in place, we run a four-family bake-off — logistic regression, RBF-SVM, Random Forest (Breiman, 2001), and XGBoost — under TimeSeriesSplit (n_splits = 5) so no fold trains on the future. All four sit close to chance (val AUCs 0.521–0.551), so the binding constraint shifts from AUC ranking to **probability spread**: NB05's quintile-based trading rule collapses to noise without it, and only RF preserves a usable spread (val $\sigma \approx 0.09$) while remaining interpretable via permutation importance. Optuna (TPE, 50 trials) tunes (max_depth = 19, min_samples_leaf = 20, max_features = 0.8); min_samples_leaf carries 55 % of the hyperparameter importance, confirming regularisation as the binding constraint. Tuning lifts val AUC 0.521 → 0.602 and top-decile precision 0.40 → 0.60. Bake-off table and Optuna log in **Appendix F**.

2.4 Three-arm trading rule and ablation (Notebook05). The tuned RF outputs a probability per event, but a probability is not a strategy. NB05 builds a three-arm refit-based ablation: arm A is BuyHold_EW50 (passive equal-weight benchmark); arm B is Tech-only (5 technicals + 11 sector dummies, refit RF with the same Optuna hyperparameters); arm C is LLM+Tech (B + 6 LLM signals, the full model). Refitting per arm is essential — it isolates the contribution of *features* from the contribution of *hyperparameter tuning*, so a B-vs-C gap cannot be confounded by either side benefiting from a better-tuned tree. Trading rules are frozen on val 2023: arm C goes long at $p \geq 0.573$, short at $p \leq 0.420$ (per-arm thresholds in **Appendix G**), holds for 10 trading days, and is evaluated against SPY. Statistical inference uses four complementary tests — Newey-West HAC t-tests on daily PnL, Memmel's Sharpe-difference z-test, a 5,000-rep block bootstrap, and event-level Spearman ρ + Mann-Whitney U on realised CAR (+2, +21). Daily-PnL tests have weak power on ~250 days/year; the event-level tests, which use ~360 independent observations, are the more credible inference layer.

3.Result

3.1 LLM signals are produced faithfully and economically meaningful. Notebook02 scores 1,496 / 1,520 sessions (98.4 %) and emits 3,529 evidence quotes, every one citing a verifiable chunk_id — **100 % hit rate, zero hallucinations**. On train 2018–2022 (n = 951), every signal correlates with realised 10-day excess-of-SPY return in the expected direction (sentiment_num $\rho = +0.107$, evasion $\rho = -0.067$; composite $\rho = +0.113$). Total cost: 5.1M tokens / 1,523 calls / ≈ \$30–50 (**Appendix D.5**).

3.2 CAR labels are well-balanced and span four event horizons. Notebook03 produces 1,480 valid (α, β, σ_e, CAR) tuples. Class balance sits at 0.51–0.54 across all four periods. Sector-level β is economically sensible (Utilities 0.30 → InfoTech 1.39; NVDA 2.04 at the high tail). After tail filter, 537/108 train/val rows remain; test sets unfiltered for backtesting. Distributions in **Appendix E**.

3.3 Random Forest wins on probability spread, not on AUC. Bake-off val AUCs cluster at chance (logreg 0.551, SVM 0.529, RF 0.521, XGB 0.543); RF wins on probability spread alone (val $\sigma \approx 0.09$, vs near-constant for linear models). Optuna lifts val AUC to **0.602** and top-decile precision to **0.60**; min_samples_leaf carries 55 % of hyperparameter importance, confirming regularization as the binding constraint. Cross-block correlation Tech↔LLM max $|\rho| = 0.14$ — **near orthogonal**, validating that the B-vs-C ablation can isolate genuine LLM contribution. Full tables in **Appendix F**.

3.4 Trading: regime-dependent edge, biggest in volatile 2025

Headline three-arm metrics (Notebook05 §6, two test years, no filter):

Period	Arm	Sharpe	Annual Ret	Vol	MDD	IR
2024 (bull)	BuyHold_EW50	**2.07**	23.7 %	10.6 %	−5.6 %	−0.24
	Tech-only	0.30	1.1 %	4.0 %	−3.4 %	−1.56
	LLM+Tech	0.43	1.6 %	3.8 %	−3.7 %	−1.53
2025 (volatile)	BuyHold_EW50	0.86	13.7 %	16.5 %	−17.1 %	−0.62
	Tech-only	−0.13	−0.5 %	3.6 %	−3.4 %	−0.94

	LLM+Tech	**0.95**	3.4 %	3.6 %	**−3.9 %**	−0.72

In 2024 BuyHold dominates because the universe pre-selects winners; LLM+Tech still beats Tech-only by +0.13 Sharpe. In 2025 the picture inverts: LLM+Tech matches BuyHold on Sharpe (0.95 vs 0.86) at one-quarter the drawdown (−3.9 % vs −17.1 %) and beats Tech-only by **+1.07 Sharpe** — the language layer earns its keep when realized volatility climbs. Curves in **Appendix A.4.3**.

3.5 Significance: weak at daily level, robust at event level, concentrated in evasion. Daily-PnL inference is honestly weak. Newey-West HAC t-tests (lag 5) on paired LLM+Tech minus Tech-only daily returns return $p = 0.93$ (2024) and $p = 0.45$ (2025); Memmel's Sharpe-difference z-test gives $p = 0.91$ and $p = 0.40$; a 5,000-rep block bootstrap on the 2025 ΔSharpe of +1.07 yields a 95 % CI of [−2.20, +3.18] — containing zero. With ~250 trading days per regime and substantial autocorrelation, daily tests do not have power to reject the null.

Event-level inference, with ~360 independent observations, is significant. Pooled across 2024–2025, the Spearman correlation between the model's predicted probability and realised **CAR (+2, +21)** is **$\rho = 0.091$, $p = 0.042$ (one-sided)**. Mean CAR for predicted-long events is +1.43 %, for predicted-short events −0.22 %, a long-minus-short spread of **+1.66 %** that is significant under a one-sided Mann-Whitney U test (**$p = 0.029$**). The per-year split is consistent with §3.4: 2024 has stronger ranking ($\rho = 0.110$) but flat Sharpe (small magnitudes); 2025 has weaker ranking ($\rho = 0.073$) but strong Sharpe (large magnitudes when the model is right).

Feature attribution attributes the lift to a small number of features. OOS permutation importance assigns **36.6 %** of the total ΔAUC to the LLM block (vs 47.0 % Tech, 16.4 % Sector); impurity-based importance gives only 11.7 %, a 3× gap consistent with the well-known impurity bias toward higher-cardinality features under heterogeneous feature scales. The block ablation confirms partial non-overlap: dropping LLM costs 0.030 val AUC, about 80 % of the cost of dropping Tech (0.038). The drop-one ablation is the sharpest finding: of the six LLM signals, only `evasion` carries individual lift (drop-one Δ = −0.021 AUC, ≈ 72 % of the full LLM block contribution); the other five show small positive deltas when dropped, meaning they are partly redundant or noisy. This is the most actionable result for prompt redesign in §5: a tighter `evasion`-focused prompt would likely match current performance at lower API cost. Full significance tables and ablation tables are in Appendix C.4–C.6 (significance) and **Appendix G** (ablation).

4.Discussion

The LLM-augmented model performs as hypothesised, with caveats. The LLM signal layer adds economically meaningful information beyond price, supporting evaluation as an event-driven decision-support tool rather than a fully automated fund. Strict anti-leakage discipline — train/val tail filtering only, val 2023 threshold freeze, refit per arm — gives reasonable confidence the OOS comparison is fair.

The strongest evidence is at the event level. Pooled Spearman $\rho = 0.091$ ($p = 0.042$) and the Mann-Whitney long-minus-short spread of +1.66 % ($p = 0.029$) confirm that predicted probabilities meaningfully sort realised CAR. The 2025 LLM+Tech Sharpe of 0.95 with MDD −3.9 % vs BuyHold −17.1 % is economically meaningful, even though daily - PnL Memmel and bootstrap tests cannot reject the null at 5 % — daily significance simply lacks power on ~250 days.

Feature attribution is a credible result, not a marketing claim. Permutation importance attributes 36.6 % to the LLM block (vs 11.7 % under the well-known impurity bias). Drop-one ablation reveals that only evasion carries individual lift ($\Delta = -0.021$ AUC, ≈ 72 % of the block); the other five LLM signals are partly redundant or noisy. This is honest about which language signals work — most do not.

Three caveats. (i) Signal stability is regime-dependent: 2024 LLM permutation importance is positive and event-level ranking strong; 2025 ranking weakens but the strategy Sharpe improves, likely capturing crisis-conditional information. (ii) The backtest assumes zero transaction costs, slippage, and borrow fees — modelling these would compress all three arms' returns. (iii) The LLM signal layer uses a single fixed prompt across 2018–2025 and may drift if rerun under a different model version; 50 large-caps for two test years are a small sample for any general claim.

5.Future Development and Expansion

The ablation in §3.5 points to a clean next step: rather than asking GPT-4o for six general scores, future work should redesign the prompt around the one signal that carried 72 % of the LLM block's lift — **evasion** — and decompose it into topic-aware diagnostics. The new prompt (**Appendix H.1**) replaces the global evasion score with four targeted questions: did management directly answer the analyst's question, did the answer migrate from numbers to qualitative language, was risk framed as transitory or structural, and which topic was deflected. Three new retrieval queries (**Appendix H.2**) target margin, demand, and guidance evasion specifically, replacing the current generic "margin pressure cost inflation" query. This converts a single scalar into an interpretable feature vector portfolio managers can actually act on.

A second extension is to move from a single-prompt LLM to a **multi-agent pipeline** built on Claude Code's agent + skill + MCP stack (**Appendix I**). Four specialised agents — *Retriever*, *Q&A-Analyst*, *Cross-Quarter-Comparer*, *Aggregator* — each carry their own system prompt, tool budget, and MCP connections (WRDS, FactSet, internal price store). This makes task decomposition explicit rather than hoping a single prompt juggles all four roles, and the per-agent traces become the audit log a compliance reviewer needs.

A third extension treats the six LLM signals as a **panel rather than i.i.d. events**. The current pipeline scores each call independently, discarding the trajectory information that distinguishes a one-off cautious quarter from a four-quarter slide into structural evasion — exactly the *transitory vs. structural* distinction raised at the prompt level. We propose stacking the signals into a ticker × quarter panel and fitting a small state-space model (HMM with 3–4 latent regimes) to extract a per-firm hidden state at every event date. The state — and its transition probability into a *deteriorating* regime — becomes a new feature for the downstream RF, alongside the raw current-quarter signals. The 50 × ~30 panel is small enough that an HMM is the right complexity choice (a GRU would overfit); the state, once trained, is interpretable as a coarse management-stance regime, which is what a portfolio manager actually wants to track between earnings dates. Full design in **Appendix J**.

6.Team Contribution

We contributed equally; every notebook was reviewed by both of us before being merged, and the report was drafted in alternating passes.

Zhankai Zhang owned the language side. In NB02 he built the RAG infrastructure — stream-aware 200-word chunking over 1,520 transcripts, the cached `text-embedding-3-large` retrieval layer, and the six-signal GPT-4o prompt with the JSON-schema and citation contract (validated at 100% chunk-id hit rate, 5.1M tokens across 1,523 calls). In NB03 he constructed the market-model CAR labels: [−120, −20] OLS estimation, four event-window horizons, and the tail filter applied to train/val only. He drafted §2.1–§2.2, §3.1–§3.2, §4, §5, and Appendices D, E, H, I, J.

Yixuan Wang owned the data and modelling side. In NB00–NB01 she built the WRDS ETL (zip parsing, ticker fuzzy-matching, the four parquet contracts every downstream notebook depends on) and the exploratory analysis. In NB04 she ran the four-model bake-off under `TimeSeriesSplit (n_splits=5)` and the Optuna RF tuning that lifted val AUC from 0.521 to 0.602 (TPE, 50 trials; final config `max_depth=19, min_samples_leaf=20, max_features=0.8`). She drafted §1, §2.3–§2.4, §3.3, and Appendices B, C, F, G.

The **NB05 three-arm ablation backtest** (headline metrics, Newey-West + Memmel + bootstrap, event-level rank tests, permutation importance, block + drop-one ablation) and the shared `src/llm_agent/` library were implemented jointly. The final report and appendix cross-references were edited together across multiple review rounds.

References

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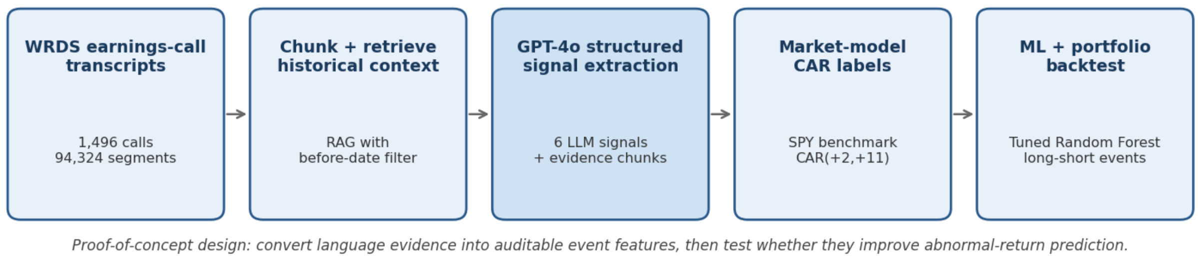
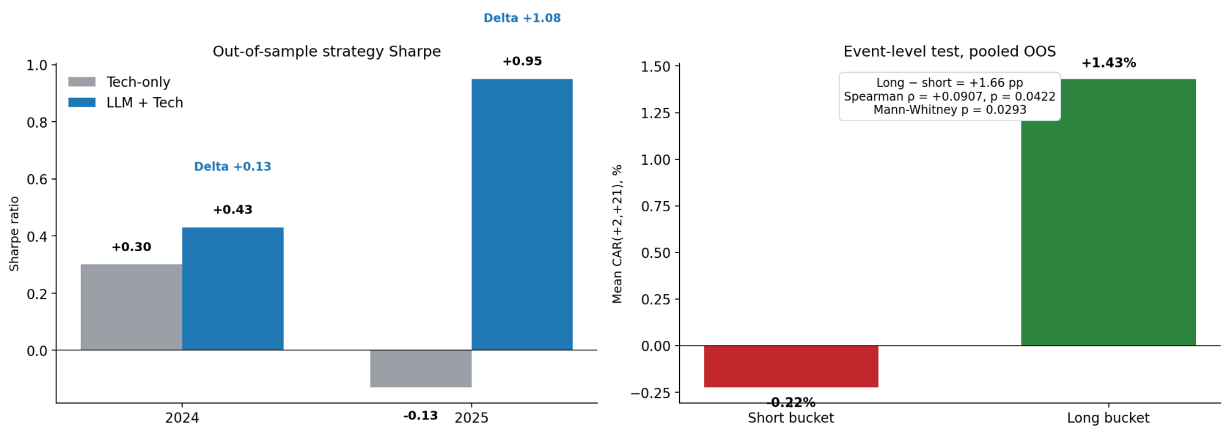
Data provenance: The project notebooks contain WRDS earnings-call transcripts, daily adjusted equity prices and the SPY market benchmark. Reported statistics are taken from the provided local project outputs:....

Appendix A. Figures

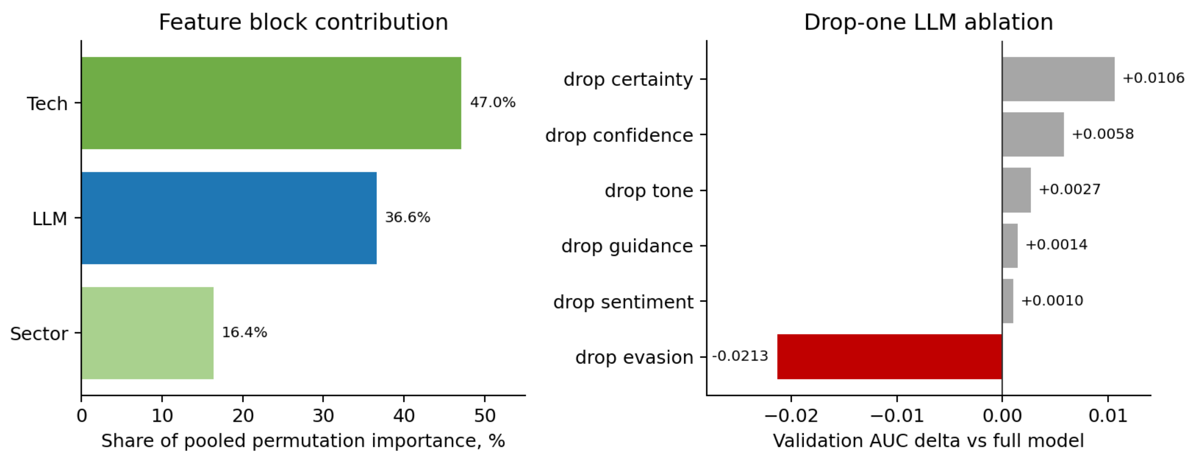
Appendix A.2.1. *Proposed LLM-augmented Event Study Pipeline and Data Flow.*



Appendix A.4.1. *Core out-of-sample results: Strategy Sharpe improvement and pooled event-level CAR separation.*



Appendix A.4.2. *Feature-attribution evidence: the LLM block carries 36.6% of pooled out-of-sample permutation importance, but the contribution is concentrated in a single feature (evasion). Drop-one ablation confirms that no other LLM signal individually moves validation AUC.*



Appendix A.4.3 Three-arm backtest: metrics and equity curves (test 2024 / 2025)

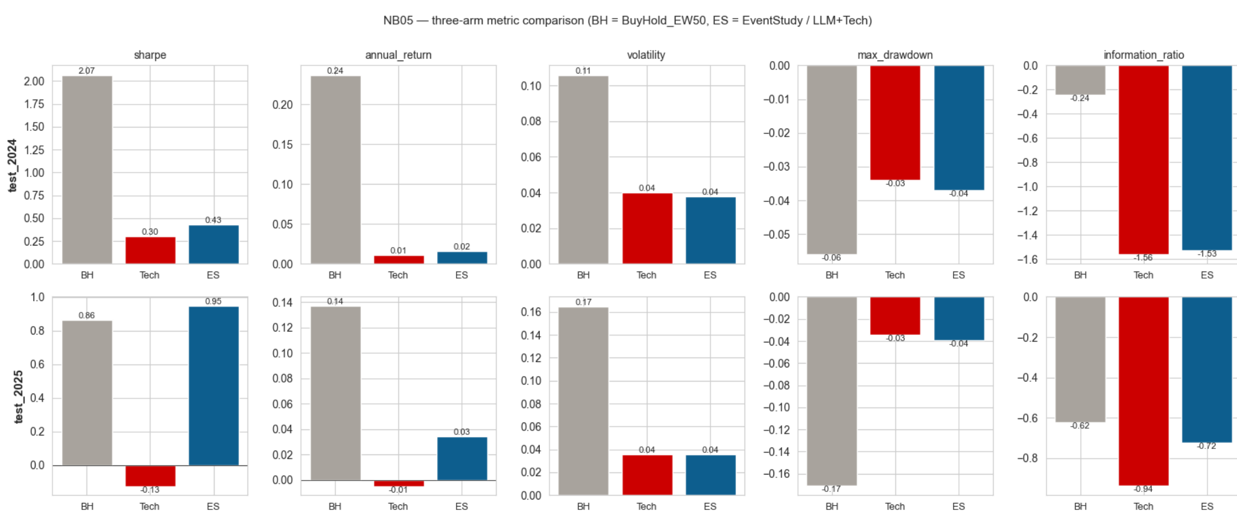


Figure A.4.3.1 — Metric grid. Five metrics (Sharpe, annual return, volatility,

max drawdown, information ratio) × two test years × three arms (BH = BuyHold_EW50, Tech, ES = LLM+Tech). Source: NB05 §6.

Top row (2024, bull): BH dominates on level (Sharpe 2.07); ES edges Tech on Sharpe and IR at near-identical vol. Bottom row (2025, volatile): the ranking inverts — ES matches BH on Sharpe (0.95 vs 0.86) at one-quarter the drawdown, while Tech turnsnegative.

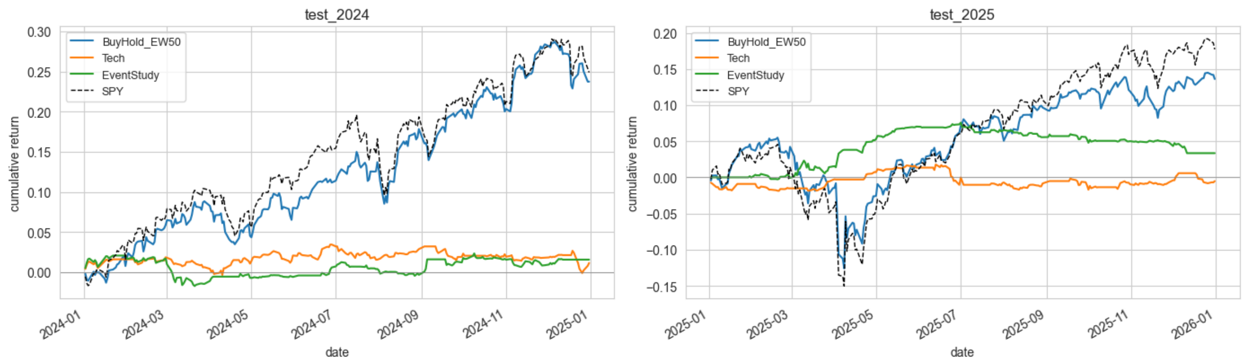


Figure A.4.3.2 — Cumulative equity curves. Two panels (2024, 2025), each with the three arms plus SPY as reference. Source: NB05 §6.

The 2025 panel is the central evidence for the "regime-dependent edge" claim: ES (green) opens a lead through April–June while SPY draws down ~15 %, then holds the gain flat through year-end as BH catches up on the recovery.

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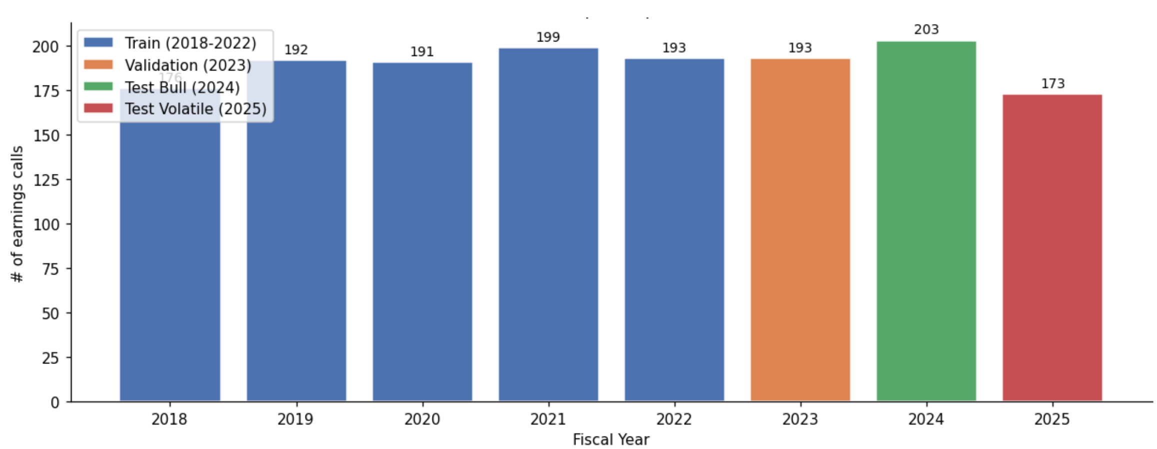
Appendix B: Exploratory Data Analysis

The cleaned dataset covers 50 large-cap US equities from 2018 Q1 to 2025 Q2: 1,520 earnings-call transcripts (median ~14k tokens each, ~88 segments split between prepared remarks and Q&A) and matched daily adjusted prices for all 50 tickers plus the SPY benchmark. 24 sessions are missing from the panel (98.4 % coverage), and a further 16 drops at the labelling stage (NB03) for insufficient pre-event estimation history. The four time-ordered splits — train (2018–2022, n = 951), val (2023, n = 193), test 2024 (n = 203), test 2025 (n = 173) — preserve the chronological ordering required for the anti-leakage protocol.



Appendix C: Model Diagnostics and Robustness Checks

This appendix introduces three diagnostic procedures used to audit the LLM-augmented event-study model. Random Forest predicts the probability of positive abnormal returns. We then test whether these predictions align with realised abnormal returns and whether performance degrades when key features are removed.

C.1 Permutation Importance — A Stethoscope

Permutation Importance is a straightforward way to quantify how important a feature is: randomly shuffle only that feature, keeping all other data unchanged, and measure the resulting drop in predictive accuracy. This is suitable for a tree model because the standard Random Forest impurity importance tends to favor continuous or high-cardinality features, even if their actual incremental value is relatively small (Breiman, 2001; Strobl et al., 2008). Based on pooled out-of-sample events and AUC as metrics, the technical block accounts for 47.05% of permutation importance, the LLM block accounts for 36.56%, and sector controls account for 16.40%. Under the impurity-based ranking, by contrast, the LLM block is only 11.70%. Evasion is the only LLM feature that is clearly performing work at the individual level and ranks second overall, only behind log dollar volume. The result is not that all language scores are useful, but rather that management evasion contains information not included in the market features. *(See Appendix A.4.2 for the visualization.)*

C.2 Event-Level Rank Tests — A Stress Test

This procedure tests whether model scores are indeed correlated with subsequent abnormal returns. The academic foundation is the event-study methodology; realised CAR isolates the firm-specific price change following an event by deducting normal market co-movements (Brown and Warner, 1985; MacKinlay, 1997). Due to the small size of the out-of-sample window for daily PnL tests, this section uses non-parametric event-level rank evidence. On pooled 2024–2025 observations ($n = 363$), the Spearman correlation between predicted probability and realised CAR(+2, +21) was $\rho = +0.0907$, and the one-sided p-value was 0.0422. The bucket test points are in the same direction: the mean CAR of the long bucket is +1.4310%, and the mean CAR of the short bucket is −0.2246%; the difference is +1.6556 percentage points, and Mann-Whitney $p = 0.0293$. Note that the model was trained on CAR(+2, +11), while the diagnostic tests use a longer CAR(+2, +21) window. The positive pooled relation therefore supports the external validity of the training label rather than confirming it. *(See Appendix A.4.1 for the visualization.)*

C.3 Feature Ablation — A Surgical Test

This procedure re-trains the model after excluding selected feature groups or individual LLM features. Based on the wrapper-style feature evaluation logic, a feature is deemed useful if the model performs worse without that information (Kohavi and John, 1997; Guyon and Elisseeff, 2003). The block ablation study shows that the LLM block has a positive marginal effect; the AUC of the full model is 0.6012, and excluding all LLM variables reduces it to 0.5713, a drop of 0.0298. Dropping the technical features is worse (AUC 0.5633), and removing the sector controls only raises the AUC slightly to 0.6036. The drop-one LLM experiment shows that without evasion, validation AUC drops to 0.5799 — 0.0213 below the full model. On the other hand, removing certainty, confidence, tone, guidance, or sentiment does not harm and often slightly increases the AUC. A cautious interpretation is that evasion is the reliable LLM signal and that the other language variables are redundant or noisy. This motivates the future-work direction of replacing the six general LLM scores with topic-specific Q&A evasion features (Section 5). *(See Appendix A.4.2 right panel for the visualisation.)*

C.4 Newey-West HAC t-tests on daily PnL (lag = 5)

Generated by NB05 §8.

Period	Series	Mean daily (bps)	NW t-stat	NW p
2024	BuyHold_EW50	8.679	2.060	0.039
2024	Tech-only	0.483	0.303	0.762
2024	LLM+Tech	0.640	0.394	0.694
2024	LLM+Tech − Tech (paired)	0.158	0.085	0.932
2025	BuyHold_EW50	5.640	0.954	0.340
2025	Tech-only	−0.180	−0.115	0.908
2025	LLM+Tech	1.345	0.902	0.367
2025	LLM+Tech − Tech (paired)	1.525	0.760	0.448

C.5 Memmel Sharpe-difference z-test + block-bootstrap CI (5,000 reps, block=10)

Period	ΔSharpe	Memmel z	Memmel p	Bootstrap 95 % CI	CI excludes 0?
2024	0.125	0.11	0.913	[−2.206, +2.303]	No
2025	1.073	0.85	0.395	[−2.196, +3.184]	No

C.6 Event-level Spearman ρ + Mann-Whitney U on CAR(+2,+21)

Window	n events	n long	n short	Spearman ρ	ρ p (1-sided)	Long mean CAR	Short mean CAR	Long−Short	MW p (1-sided)
Pooled 2024–2025	363	67	87	**+0.0907**	**0.0422**	+1.43 %	−0.22 %	**+1.66 %**	**0.0293**
2024	198	33	50	+0.1099	0.0617	+1.23 %	−0.99 %	+2.22 %	0.0585
2025	165	34	37	+0.0730	0.1759	+1.63 %	+0.81 %	+0.82 %	0.2324

Appendix D: RAG-LLM Implementation Detail

Generated by NB02. Supports §2.1 and §3.1.

D.1 System and user prompts (verbatim)

System:

```
You are an equity analyst evaluating an earnings call.
Answer ONLY using the provided excerpts. If the excerpts do not support a
field, output `null` or 'none'. Cite the chunk_id of every quote you use.
Output a single JSON object — no prose, no markdown fences.
```

User template:

```
Below are retrieved excerpts from {ticker}'s {year} Q{quarter} earnings call,
plus excerpts from prior quarters of the same company.

Return JSON with these fields:
{
  "sentiment":   "bullish" | "neutral" | "bearish",
  "confidence":  float in [0,1],
  "certainty":   int in [1,10],
  "evasion":     int in [1,10],
  "guidance_change": "raise" | "maintain" | "lower" | "none",
  "tone_vs_prior_quarter": "more_cautious" | "similar" | "more_confident",
  "evidence": [{"chunk_id": "...", "quote": "..."}, ...]   // ≥2 entries
}

Context:
{context}
```

D.2 Retrieval queries (three fixed per session)

1. forward guidance demand outlook
2. margin pressure cost inflation
3. capital allocation buyback dividend

Anti-leakage filter: only chunks whose parent session has mostimportantdate < event_date are eligible from prior quarters.

D.3 Hyperparameters

Setting	Value
Embedding model	text-embedding-3-large (3,072 dim)
Embedding batch size	96 texts/call
Chunk size / overlap	200 / 50 words

Retrieval top-k	5 per query × 3 queries
LLM model	gpt-4o
LLM temperature	0.0
LLM max_tokens	1024
Response format	json_object
Disk flush cadence	every 25 sessions

D.4 Signal validation on training set (2018–2022, n = 951)

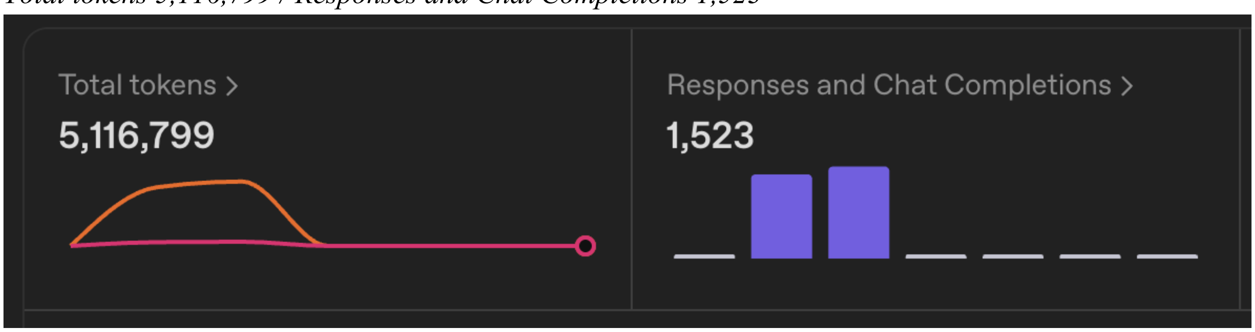
Pearson correlation with realised 10-day excess-of-SPY return:

Signal	ρ
sentiment_num	+0.107
guidance_num	+0.095
tone_num	+0.065
certainty	+0.063
confidence	+0.061
evasion	−0.067
Composite	**+0.113**

D.5 OpenAI usage (full pipeline run)

Metric	Value
Total tokens consumed	**5,116,799**
Chat completions (gpt-4o)	**1,523**
Sessions scored	1,496 / 1,520 (98.4 %)
Evidence quotes emitted	3,529 (avg 2.4 per call)
Hallucinated chunk_ids	0 (100 % hit rate)
Approximate cost	\$30–50

Total tokens 5,116,799 / Responses and Chat Completions 1,523



D.6 Output schemas

data/signals/llm_signals.parquet — 1 row per session, 15 cols: session_id, ticker, fiscal_year, fiscal_quarter, event_date, sentiment, sentiment_num, confidence, certainty, evasion, guidance_change, guidance_num, tone_vs_prior_quarter, tone_num.
data/signals/evidence.parquet — 1+ rows per session: session_id, chunk_id, quote, hit (where hit=True confirms the cited chunk was in the retrieval pool).

Appendix E: CAR Label Construction Diagnostics

Generated by NB03. Supports §2.2 and §3.2.

E.1 Event window definitions

Column	Window	Use
car_2_3	[+2, +3]	short-window robustness
car_2_6	[+2, +6]	mid-window robustness
car_2_11	[+2, +11]	**primary training target (NB04)**
car_2_21	[+2, +21]	longer-window PEAD evaluation (NB05 §9)

E.2 Sample sizes and class balance per period

Period	Total events	After tail filter	y=1 rate (full)	y=1 rate (tail)
Train 2018–2022	925	537	0.508	0.507
Val 2023	192	108	0.526	0.500
Test 2024	198	120	0.530	0.533
Test 2025	165	98	0.539	0.449

Tail filter rule: $|CAR| > 0.5 \cdot \sigma\_e \cdot \sqrt{10}$. Applied to train/val only; test sets unfiltered.

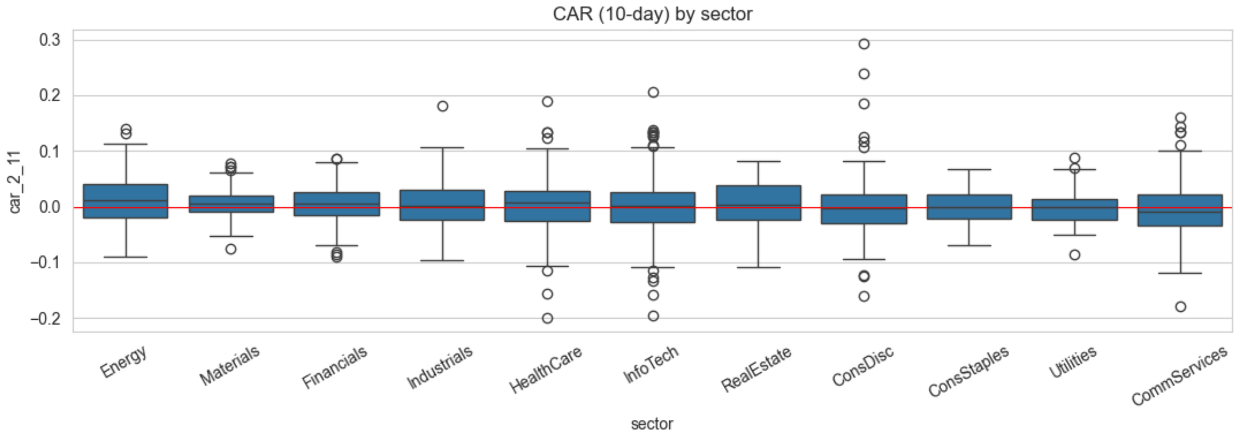
E.3 CAR descriptive statistics (n = 1,480)

Stat	car_2_3	car_2_6	car_2_11	car_2_21
mean	+0.0006	+0.0009	+0.0015	+0.0009
std	0.0237	0.0330	0.0449	0.0647
min	−0.1038	−0.1245	−0.1992	−0.3492
25 %	−0.0122	−0.0188	−0.0254	−0.0350
median	0.0000	−0.0001	+0.0014	−0.0008
75 %	+0.0127	+0.0201	+0.0264	+0.0379
max	+0.2170	+0.1794	+0.2926	+0.3131

E.4 Sector-level β (market model on SPY)

Sector	Avg β	Ticker (high)	β	Ticker (low)	β
InfoTech	1.39	NVDA	2.04	DUK	0.29
CommServices	1.08	TSLA	1.89	SO	0.31
Financials	1.06	AVGO	1.53	JNJ	0.40
Industrials	1.04	NOW	1.45	PG	0.40
ConsDisc	0.95	META	1.42	WMT	0.42
Materials	0.85				
Energy	0.79				
RealEstate	0.71				
HealthCare	0.53				

ConsStaples	0.50
Utilities	0.30



Appendix F: Model Bake-off and Optuna Tuning

Generated by NB04. Supports §2.3 and §3.3.

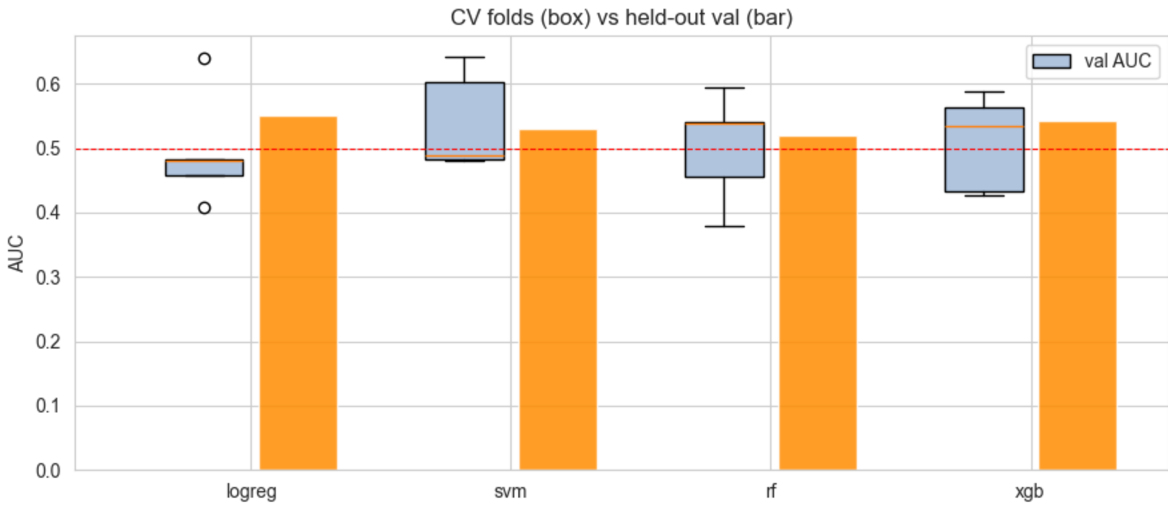
F.1 Feature inventory (22 features)

Block	Features	Count
Tech	rsi_7, macd_hist, mom_5d, vol_10d, log_dollar_vol	5
Sector	11 GICS dummies (sec_CommServices, sec_ConsDisc, …, sec_Utilities)	11
LLM	sentiment_num, confidence, certainty, evasion, guidance_num, tone_num	6

Feature engineering anti-leakage: technicals computed strictly with < on event_date.

F.2 Bake-off results (TimeSeriesSplit n_splits=5, scoring=roc_auc)

Model	CV AUC mean	CV AUC std	Val AUC	Val Acc	Val Top-10 % Precision
logreg	0.494	0.078	0.551	0.519	0.7
svm	0.540	0.069	0.529	0.519	0.6
rf	0.502	0.075	0.521	0.546	0.4
xgb	0.509	0.067	0.543	0.565	0.3

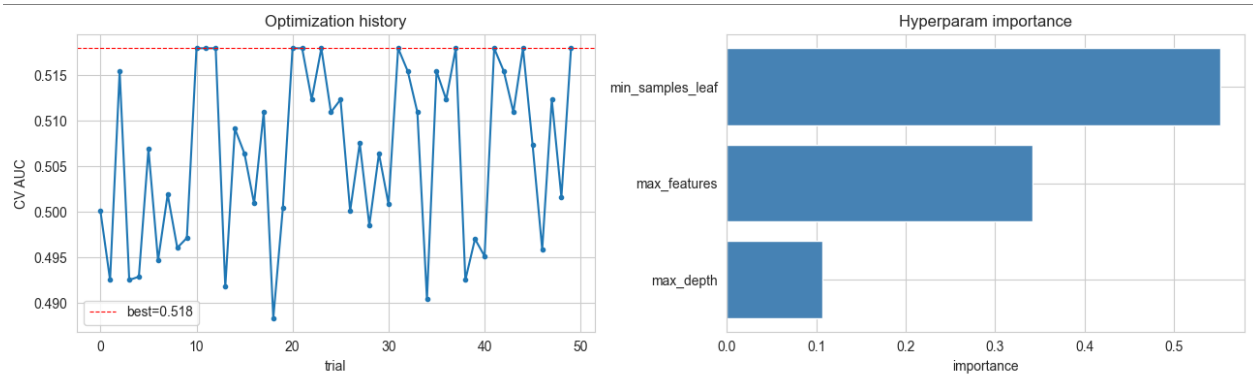


Selection criterion: probability spread (val σ ≈ 0.09 for RF only); see §2.3 / §3.3.

F.3 Optuna RF tuning (TPE sampler, 50 trials, MedianPruner)

Search space: max_depth ∈ [3, 20], min_samples_leaf ∈ [1, 20], max_features ∈ {sqrt, 0.3, 0.5, 0.8}. Best params: max_depth = 19, min_samples_leaf = 20, max_features = 0.8.
Hyperparameter importance:

Hyperparameter	Importance
min_samples_leaf	0.55
max_features	0.34
max_depth	0.10



F.4 Tuned RF performance

Metric	Base RF	Tuned RF	Δ
CV AUC	0.502	0.518	+0.016
Val AUC (2023)	0.521	**0.602**	+0.081
Val Acc	0.546	0.630	+0.084
Val Top-10 % Precision	0.40	**0.60**	+0.20

F.5 Cross-block feature correlations

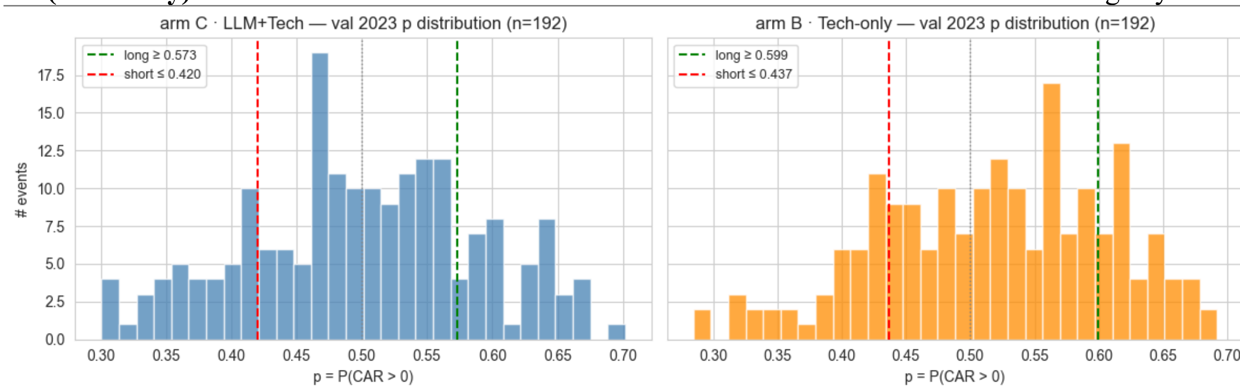
- Tech ↔ Tech: max |ρ| = 0.76 (rsi_7 ↔ mom_5d)
- LLM ↔ LLM: max |ρ| = 0.87 (confidence ↔ certainty)
- Tech ↔ LLM: max |ρ| = 0.14 — **near orthogonal** (validates the B-vs-C ablation)

Appendix G: Trading Rule and Ablation Detail

Generated by NB05 §7 / §10 / §11. Supports §2.4 and §3.5.

G.1 Frozen trading thresholds (val 2023, n = 192)

Arm	Long threshold (p ≥)	Short threshold (p ≤)	Holding period
C (LLM+Tech)	0.5729	0.4203	10 trading days
B (Tech-only)	0.5990	0.4370	10 trading days



Predicted-probability distributions on val 2023:

Arm	Mean	Std	Min	Q1	Median	Q3	Max
C	0.499	0.091	0.246	0.436	0.503	0.568	0.740
B	0.512	0.088	0.241	0.451	0.518	0.576	0.742

Figure G.1: val-2023 probability histograms for arms B and C with frozen thresholds overlaid

G.2 Permutation importance (pooled OOS, n = 363)

Block	Δ AUC	% of total	Impurity share	Impurity %
Tech	0.0073	47.0 %	0.848	84.8 %
Sector	0.0025	16.4 %	0.035	3.5 %
LLM	0.0056	**36.6 %**	0.117	**11.7 %**

Per-year ΔAUC by block:

Block	2024	2025
Tech	0.0296	0.0225
Sector	0.0023	0.0033
LLM	+0.0177	−0.0021

G.3 Block ablation (refit RF, val AUC)

Configuration	n features	Val AUC	Δ vs full	Top-decile precision
Full (Tech + Sector + LLM)	22	0.6012	—	0.5
Drop Tech	17	0.5633	**−0.0379**	0.7
Drop Sector	11	0.6036	+0.0024	0.5
Drop LLM	16	0.5713	**−0.0298**	0.7
LLM only	6	0.5430	−0.0581	0.7
Tech only	5	0.5957	−0.0055	0.6

G.4 Drop-one LLM-feature ablation (refit RF, val AUC)

Drop	n features	Val AUC	Δ vs full
(none — full model)	22	0.6012	—
sentiment_num	21	0.6022	+0.0010
confidence	21	0.6070	+0.0058
certainty	21	0.6118	+0.0106
evasion	21	**0.5799**	**−0.0213**
guidance_num	21	0.6025	+0.0014
tone_num	21	0.6039	+0.0027

Only evasion carries individual lift; it accounts for ≈ 72 % of the full LLM-block contribution from G.3.

Appendix H: Redesigned Prompt and Retrieval Queries (proposed)

H.1 Topic-aware system prompt

Replaces the v1 single-scalar evasion field with a per-topic diagnostic block. JSON-only output, citation required, null allowed when excerpts do not support a field.

You are an equity analyst evaluating an earnings call. Answer ONLY using the provided excerpts. If the excerpts do not support a field, output `null`. Cite the chunk_id of every quote you use. Output a single JSON object — no prose, no markdown fences.

For EACH of the four topics {margin, demand, guidance, capital_allocation},return a sub-object with the following diagnostics:

```
"directness":        int in [1,10]   // did management answer the question
                                     // posed, or pivot away?
"register_shift":    "numeric_to_qualitative" | "qualitative_to_numeric"
                     | "stable" | "n/a"
"risk_framing":      "transitory" | "structural" | "mixed" | "n/a"
"deflection_target": str | null      // if directness <= 4, name the topic
                                     // management redirected the answer to
```

```
  "evidence":          [{"chunk_id": "...", "quote": "..."}]   // ≥1 entry
```

H.2 Topic-aware retrieval queries

The current single query "margin pressure cost inflation" is replaced by a matrix of {topic × stance} queries, run independently and deduplicated by chunk_id before being concatenated into the LLM context.

Topic	Query A (factual)	Query B (forward)
margin	gross margin operating margin compression	margin outlook pricing power next quarter
demand	order book backlog bookings unit volume	demand visibility customer commitments next year
guidance	full-year guidance range raised lowered	implied second-half exit-rate trajectory
capital_allocation	buyback authorisation dividend increase	capex priorities M&A pipeline cash deployment

Top-3 chunks per query × 8 queries → up to 24 chunks pre-dedup, vs the current top-5 × 3 = 15. Cost increase ≈ 1.6× tokens per call; mitigated by reducing max_tokens on the response from 1024 to 768 (the new schema is more compact).

H.3 Migration plan

1. Run H.1 / H.2 on the 2018–2022 train period only; freeze prompt v2.
2. Compare per-topic directness and register_shift to v1 evasion on shared sessions — confirm the new vector spans the old scalar.
3. Re-train RF on (5 technicals + 11 sectors + 4 topics × 4 diagnostics); compare val AUC and pooled OOS Sharpe against the v1 LLM+Tech arm.
4. If v2 wins on val 2023, re-run NB05 §6–§11 end-to-end.

Appendix I: Multi-Agent Architecture (proposed)

I.1 Agent roster

Agent	Role	System prompt focus	Tools / MCP
Retriever	Pull call excerpts under before_date filter	Return the smallest set of chunks that span all four topics; prefer Q&A turns over prepared remarks.	embeddings.search, wrds.session_lookup (MCP)
Q&A-Analyst	Score directness, register shift, risk framing per topic	H.1 prompt, restricted to current-quarter chunks	none (LLM only)

Cross-Quarter-Comparer	Score tone_vs_prior_quarter and detect topic drift across the last 4 calls	Compare the current call's stance on each topic to the same topic in t−1, t−2, t−3, t−4.	wrds.session_history (MCP)
Aggregator	Combine outputs into the final feature vector + evidence pack	Reconcile contradictions; flag any topic with <2 evidence quotes.	none

I.2 Tool / skill bindings

Skill	Owner agent	Purpose
transcript_chunk_search	Retriever	Cosine top-k over cached text-embedding-3-large vectors
factset_consensus_pull	Q&A-Analyst	Compare guidance language to sell-side consensus deltas
price_reaction_lookup	Aggregator	Attach realised CAR(+2,+11) to evidence pack for human review

MCP servers: wrds-mcp (read-only on earnings_segments), factset-mcp (consensus + revisions), internal-prices-mcp (CRSP daily, SPY benchmarks).

I.3 Cost and latency budget (proposed cap per call)

Agent	Model	Max tokens in / out	Wall-clock target
Retriever	claude-haiku-4-5	4k / 2k	< 3 s
Q&A-Analyst	claude-opus-4-7	12k / 1k	< 8 s
Cross-Quarter-Comparer	claude-sonnet-4-6	16k / 1k	< 6 s
Aggregator	claude-haiku-4-5	6k / 1k	< 3 s
Total	mixed	~38k / 5k	< 20 s/call

Estimated full-pipeline cost for 1,520 calls: ≈ \$120–180 (vs the current \$30–50 single-prompt GPT-4o run), justified only if v2 lifts pooled Sharpe by ≥ 0.2 over v1 on val 2023 — the gating criterion before any production rollout.

I.4 Why multi-agent beats one big prompt

1. **Specialisation under context budget** — each agent sees only the chunks relevant to its role, so the Q&A-Analyst is not paying attention tax on cross-quarter context the Comparer handles separately.
2. **Independent evidence chains** — the Aggregator can refuse to emit a topic score if two agents disagree, giving a per-topic confidence the v1 single-prompt design cannot produce.

3. **Auditability** — each agent's input, tool calls, and output are logged separately, satisfying the "every numeric output traceable to a verbatim chunk" requirement at agent granularity rather than only at the final-JSON granularity.

Appendix J: Panel State-Space Extension (proposed)

J.1 Data layout

Reshape data/signals/llm_signals.parquet from event-level to a balanced ticker × fiscal-quarter panel:

Dim	Size	Notes
Tickers	50	config.TARGET_COMPANIES
Quarters	~30	2018Q1 – 2025Q2
Features per cell	6	sentiment_num, confidence, certainty, evasion, guidance_num, tone_num
Total cells	~1,500	matches NB02 scored sessions

Missing cells (24 skipped sessions) are forward-filled within ticker; if ≥ 2 consecutive quarters are missing, the ticker is dropped from the panel for that window.

J.2 Model

Per-ticker Gaussian HMM with K = 3 latent states (interpretable as constructive, neutral, defensive); transition matrix A is shared across tickers (pooled estimation), emission means / covariances are ticker-specific to allow for firm-level baselines (e.g. NVDA's average certainty is structurally higher than UTL's). Code sketch:

```
from hmmlearn.hmm import GaussianHMM
model = GaussianHMM(n_components=3, covariance_type="diag",
                    n_iter=200, tol=1e-3, random_state=42)
# Fit per-ticker on train period only (2018Q1 – 2022Q4)
# Decode posterior state for val + test quarters with no refit
```

Anti-leakage: HMM parameters are frozen on the train period; val 2023 and test 2024 / 2025 states come from predict_proba on the held-out quarters, never from re-fits that include them.

J.3 New features fed back into RF

Feature	Definition	Type
state_t	argmax posterior at event quarter	categorical (one-hot, 3 dummies)
state_prob_defensive	P(state = defensive | obs ≤ t)	float [0,1]
state_persistence	run-length of current state	int
transition_into_defensive	1 if state_t = defensive and state_{t-1} ≠ defensive	binary

Total: 6 new features → feature count rises from 22 to 28 in the LLM+Tech arm.

J.4 Evaluation protocol

1. Refit RF (Optuna hyperparams from NB04) on the augmented feature matrix; compare val 2023 AUC to the v1 LLM+Tech arm (baseline 0.602).
2. Re-run NB05 §6–§11 with the augmented arm; the gating criterion is **pooled OOS Sharpe lift ≥ 0.15** over v1.
3. Permutation-importance the 6 new features as a block — if the panel block carries < 5 % of pooled lift, drop it (the i.i.d. assumption was harmless).

J.5 Why HMM, not GRU

Consideration	HMM	1-layer GRU
Sample size	OK at ~1,500 cells	underpowered, high variance
Interpretability	latent state directly inspectable	requires post-hoc probing
Anti-leakage	trivial — freeze A, μ, Σ	easy to leak via shared embeddings
Compute	seconds	minutes per fold

A GRU becomes the right choice once the universe expands to Russell 1000 — until then, HMM is the parsimonious match to the data budget.